11) Figure 1 shows the Ramachandran plot for two amino acids (Proline and Glycine).

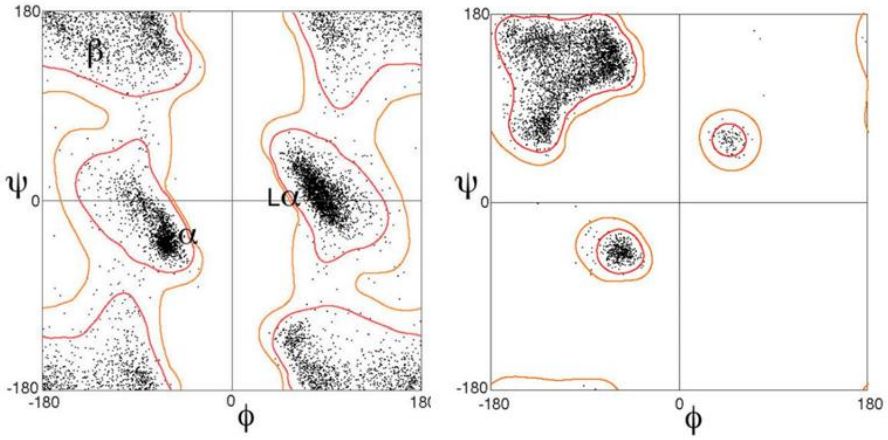


Figure 1.

(a) Identify which amino acid is proline and which is glycine, and justify.
(b) Explain what the regions marked with α and β represent.

The left Ramachandran plot corresponds to glycine, and the right plot corresponds to proline.
In a Ramachandran plot, the relative position of two consecutive amino acids is described by two dihedral angles (ϕ and ψ), and not every combination of ϕ and ψ is possible, due to steric and electrostatic hindrance; moreover, the values of the two dihedral angles are linked to the local secondary structure.
Because different amino acids have different preferred orientation (which explain their secondary structure propensity), glycine has the smallest side chain -H, so it has minimal steric hindrance and greater backbone conformational flexibility, showing a broader allowed area (including more ϕ and ψ combinations). Proline's side chain forms a ring with the backbone; it shows a more restricted distribution because its cyclic side chain constrains backbone geometry (especially restricting ϕ).

The α region represents backbone (ϕ, ψ) combinations typical of an α-helix (right-handed), α-helix angles are approximately $\phi \approx -60°$ and $\psi \approx -45°$.
The β region represents backbone (ϕ, ψ) combinations typical of β-strands/β-sheets (extended conformations).

12) The formula below can be used to describe a classical potential describing biological macromolecules. Briefly explain what each part of the formula represents.

$$U = \sum_{bonds} k_r(r - r_0)^2 + \sum_{angles} k_\theta(\theta - \theta_0)^2 + \sum_{dihedrals} C_k \cos(\varphi - \varphi_{0,k})) + \sum_{impropers} k_\alpha(\alpha - \alpha_0)^2 + \sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}} + \sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}}\right)$$

This expression is a classical force-field potential energy, written as a sum of bonded terms (bonds/angles/dihedrals/improper dihedrals) and non-bonded terms (electrostatics/van der Waals).

Bond stretching (bonded): $\sum_{\text{bonds}} k_r\,(r - r_0)^2$

Deformation of bond lengths from equilibrium. Bonds are treated as harmonic oscillators: atoms oscillate around their equilibrium average distances.

r= bond length, r_0= equilibrium bond length, k_r= bond stiffness

Angle bending (bonded): $\sum_{\text{angles}} k_\theta\,(\theta - \theta_0)^2$

Deformation of bond angles from equilibrium (harmonic approximation).

θ= bond angle, θ_0= equilibrium angle, k_θ= angular stiffness.

Dihedral (torsional) (bonded): $\sum_{\text{dihedrals}} C_k \cos(\varphi - \varphi_{0,k})$

Energy preference for rotation about a bond (controls conformations). Four consecutive atoms define three bonds and two planes; the angle between planes is the dihedral angle.

φ= dihedral angle, C_k= torsional strength, $\varphi_{0,k}$= phase (preferred offset)

Improper dihedral (bonded): $\sum_{\text{impropers}} k_\alpha\,(\alpha - \alpha_0)^2$

Constraint to maintain planarity of specific groups (e.g., rings/planar centers).

α= improper dihedral, α_0= target value, k_α= stiffness.

Coulomb electrostatics (non-bonded): $\sum_{i<j} \frac{q_i q_j}{4\pi\varepsilon_0 r_{ij}}$

Electrostatic interaction between non-bonded atom pairs. The charge represents the average charge on the atom due to the distribution of the electrons in that particular molecule.

q_i, q_j= atomic charges, r_{ij}= distance, ε_0= vacuum permittivity; $i < j$avoids double counting.

Lennard–Jones (van der Waals) (non-bonded): $\sum_{i<j} \left(\frac{A}{r_{ij}^{12}} - \frac{B}{r_{ij}^{6}}\right)$

Short-range repulsion (A/r^{12}, electron cloud overlap) and longer-range attraction(dispersion) ($-B/r^6$). It is the sum of an attractive part (quantum effect) and a repulsive part which represent the effect of electronic clouds in close proximity.

$A, B$are parameters, $i < j$ avoids double counting.

Hydrogen-bond

an empirical potential that takes into account the possibility of creation of hydrogen bonds.

C/rij^{12}: short-range repulsion; -D/rij^{10}: stabilizing attraction.

```
ATOM      1  N   ARG A  20      24.557 -38.438  -2.219  1.00 48.18           N
ATOM      2  CA  ARG A  20      24.786 -38.383  -3.724  1.00 48.20           C
ATOM      3  C   ARG A  20      26.209 -38.780  -4.233  1.00 46.86           C
ATOM      4  O   ARG A  20      26.312 -39.492  -5.209  1.00 46.96           O
ATOM      5  CB  ARG A  20      24.328 -37.035  -4.295  1.00 48.44           C
ATOM      6  CG  ARG A  20      23.791 -37.076  -5.718  1.00 49.20           C
ATOM      7  CD  ARG A  20      23.543 -35.655  -6.309  1.00 49.60           C
ATOM      8  NE  ARG A  20      22.419 -34.974  -5.669  1.00 54.70           N
ATOM      9  CZ  ARG A  20      22.453 -33.754  -5.107  1.00 55.47           C
ATOM     10  NH1 ARG A  20      23.558 -33.001  -5.109  1.00 52.88           N
ATOM     11  NH2 ARG A  20      21.345 -33.268  -4.550  1.00 55.28           N
```

(a) Explain what each of the columns represents.

(b) Explain why different carbon atoms in column C are labeled differently.

Everything following the 'ATOM' word indicates the properties of an atom in the protein.

Record type = ATOM (this line describes an atom record in a PDB file).

Atom number (atom serial index in the file).

Atom "Type"/Atom name (e.g., N, CA, C, O, CB, CG…).

Residue name (3-letter code).

Chain identifier (distinguishes different polymer chains).

Residue number (sequence index of the residue in that chain).

x coordinate (in Å).

y coordinate (in Å).

z coordinate (in Å).

Element (chemical element symbol, e.g., C, N, O).

Occupancy (fraction of time spent in that position for each atom).

B-factors (fluctuations around the average positions $Å^2$).

Because a residue contains multiple distinct carbon atoms, the PDB format must assign unique atom names so each atom can be unambiguously identified in the structure.

In amino acids, the backbone carbons have standard names:

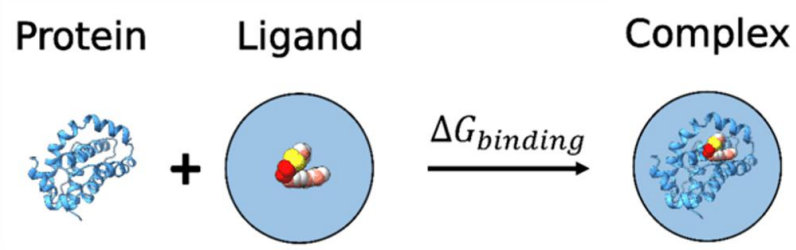
CA = α-carbon (the central carbon of the amino acid backbone)

C = carbonyl carbon of the peptide backbone

The side-chain carbons are named by position away from CA: CB, CG, CD …

Some residues (e.g., ARG) have functional group-specific naming because chemically different carbons in the side chain.

So, different carbon atoms must be labeled differently



(a) The Gibbs Free Energy is given by: $G = U + pV - TS$. Explain what the internal energy U and the entropy S are from the microscopic point of view.
(b) Define the binding Free Energy and explain what the difference in free energies between the bound and unbound states represents thermodynamically. Briefly explain what a negative value for ΔG means.

Internal energy U: The energy associated with a given microscopic configuration of the system. it is the interaction energy represented by a classical potential energy between atoms.
Entropy S: S reflects the number of accessible microscopic configurations.

The binding free energy is the free-energy difference between the bound state and the unbound state: $\Delta G_{binding}=G_B-G_U$, and can be written as $\Delta G = \Delta H - T\Delta S = G_B - G_U = (U_B - U_U) - T(S_B - S_U)$ (with $\Delta H = \Delta U + p\Delta V = \Delta U$ if no volume change). Thermodynamically, $G_B - G_U$ represents how much the free energy changes upon forming the complex, and it determines binding affinity. $\Delta G=RTlnK_D$
A negative ΔG means binding is thermodynamically favourable: the bound state has lower free energy than the unbound state at that temperature (stronger binding corresponds to smaller K_D, $K_A = 1 / K_D$).

(a) The binding free energy of a ligand (small molecule A) to a protein can be described by the formula ΔG=ΔH−TΔS. Explain what ΔG, ΔH and ΔS are in this context and how they can be interpreted at the microscopic level.
(b) Assume there is a second ligand (small molecule B) with binding affinity ΔGB<ΔGA. Explain how the two molecules will compete for binding to the protein if they are added to the solution at the same concentration.
(c) Which of the two molecules in question (b) is a better drug candidate if we consider the binding affinity alone? Motivate your answer.

Binding free energy (ΔG): the difference of Gibbs free energy between the bound and unbound states: $\Delta G = G_B - G_U = \Delta H - T\Delta S$.
Microscopically, it can be written as: $\Delta G = U_B - U_U - T(S_B - S_U)$, so it quantifies how the balance between internal energy and entropy changes upon forming the complex.
Enthalpy change (ΔH): energy changes associated with breaking and forming intermolecular interactions (hydrogen bonds, van der Waals, electrostatics) during binding, plus (usually unfavorable) energy changes due to loss of solvation of the protein and ligand.
Microscopic view: ΔH is largely about changes in the interaction energy landscape of the system upon complex formation.
Entropy change (ΔS): the change in the configurational space accessible to the protein and ligand upon binding (usually unfavorable), plus possible favorable contributions from release/reorganization of water molecules at the binding interface.
Microscopic view: entropy reflects how many microscopic configurations are accessible; at 310K, different states are accessible, so entropy matters and free energy selects the most probable state.
To compare ligands, $\Delta\Delta G = \Delta G_B - \Delta G_A$ and $\frac{K_{D,A}}{K_{D,B}} = e^{(\Delta G_A - \Delta G_B)/k_B T} = e^{-\Delta\Delta G/k_B T}$. If $\Delta G_B < \Delta G_A$, then $\Delta\Delta G < 0$, B is the better binder.
When A and B at the same concentration, B will preferentially occupy the protein binding sites, and at equilibrium, it will tend to outcompete (and effectively displace) A because the bound state with B is thermodynamically more favorable.
Ligand B is the better candidate based on binding affinity alone, because $\Delta G_B < \Delta G_A$implies stronger binding (and correspondingly a smaller K_D, since $\Delta G = RT\ln K_D$, and $K_A = 1/K_D$).

Explain the concept of the protein folding funnel and its significance in protein structure formation (10 marks). Discuss the role of entropy in the folding process (10 marks).

Levinthal's paradox: Protein cannot explore the whole configuration space to find the native state (minimum of energy).

A possible way to solve Levinthal's paradox is assuming that the energy landscape is shaped as a funnel

The funnel shape represents a bias toward the native state, with energy decreasing as the protein approaches its correctly folded structure.

Multiple folding pathways can lead to the same native state, eliminating the need to search through all possible conformations randomly.

Local minima in the funnel can act as folding intermediates or misfolded traps, but the overall downward slope drives folding toward the global energy minimum.

Folding is primarily driven by the hydrophobic effect, where hydrophobic residues collapse into the protein core to minimize unfavorable contacts with water, while hydrophilic residues remain on the surface.

Significance: the funnel model explains why proteins can fold efficiently (guided downhill landscape + multiple pathways)

In the funnel representation, the width of the funnel represents conformational entropy (number of possible states), which decreases as folding progresses and the structure becomes more ordered.

Unfolded state: many accessible conformations → high conformational entropy (wide funnel)

Folding progression: the funnel narrows → fewer accessible conformations → conformational entropy decreases (more ordered).

Native state: few accessible conformations → low conformational entropy (narrow bottom of the funnel).